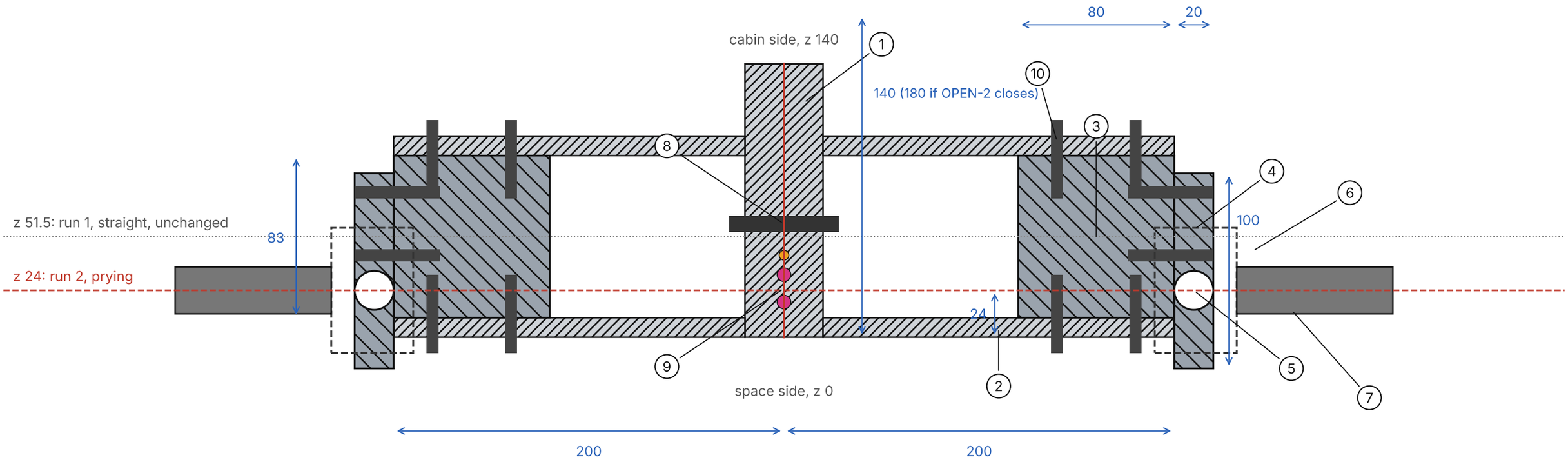
Sheet RE-03 revision B (2 Nov 2026, PROPOSAL): the T1 slice in the prying pull, side view, pinned lugs at the z 24 load line

Both pull blocks hang from pins parallel to the seam axis, so each end turns freely and the pull through the outer flange pries the joint about its cabin-side edge, as the D1 run 2 model of 2 Nov assumes. Only the z 24 M24 thread goes; everything else on RE-01 to RE-03 stays.



1 Frame A or B, 6061-T651, Rev S section as on RE-01 (140 deep; 180 if OPEN-2 closes), 200 mm grip legs.

2 Grip legs, outer flange and seat lip, 10 thick, with 13.5 mm clearance holes for the M12 leg screws (RE-01, OPEN-6).

3 Pull block, 4140 QT 28 to 32 HRC, 83 × 80 x slice length, unchanged except: the z 24 M24 thread is deleted; 4 x M12 × 1.75 tapped holes 20 deep in the pull face at z 42 and z 74, 30 mm either side of mid-length.

4 NEW lug plate F-01: 4140 QT, 80 (x) x 100 (z) x 20 thick, screwed to the pull face with 4 x M12 × 45 class 12.9 and hardened washers; it hangs 16 mm below the block's outer face. Pin hole 20 H7 at z 24, 40 mm from the lug's lower edge (2d).

5 NEW pin F-03: 20 h6 × 70, 17-4PH H900 or 4140 QT, with a shoulder and an E-clip; greased.

6 NEW fork F-02: 4140 QT, two cheeks 15 thick with a 21 mm gap, 20 H8 holes, body 60 × 60 with the M24 × 3 female thread for the existing Instron adapter; one fork each end, same for all three slice lengths.

7 Existing M24 stud and adapter to the frame's crosshead and base.

8 5/16 seam bolt at z 58 (NAS6705, preload by torque, 2.5 to 2.75 kN).

9 The two butyl cords (z 18 and 32) and the RTV bead (z 42) on frame B's joint face; the leak port between the cords.

10 M12 × 35 class 12.9 leg screws with hardened washers, through the legs into the block (RE-01-2).

Fixture design load 50 kN, half the frame's 100 kN: pin in double shear 2 × 314 mm2 × 600 MPa = 377 kN; lug tear-out 2 × 20 × 30 × 600 = 720 kN; the four lug screws carry 50 kN of pull plus the 50 kN x 34 mm = 1.7 kNm of prying on the lug (pin at z 24, screw rows at z 42 and 74) as 27 kN per screw on the far row plus 12.5 kN direct, 39 kN against a 12.9 proof of about 83 kN each.

The 550 mm article (RE-02) gets the same grip legs, two 550 mm blocks (29 kg each) and lugs so it can be pulled in prying for the gap-against-leak measurement, and a dummy hardpoint station at mid-length (the three pin holes and pockets of the flight fitting, pins fitted loose) so the seals are tested past a fitting. Expected pull to a 0.33 mm gap on the 225 slice: about 26 kN; on the 550 article about 61 kN.